

Curves and differentiation

An introduction to slopes, intercepts and some techniques for finding them

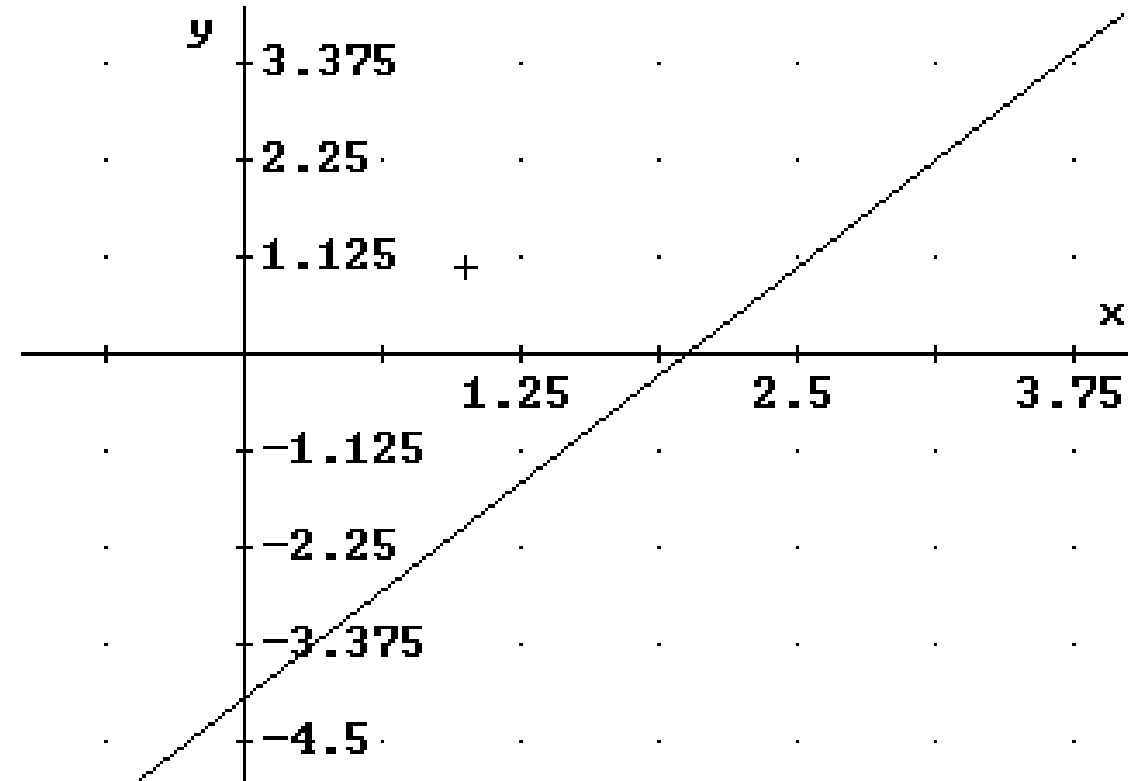
Straight lines

We define the straight line by two quantities, its slope (m) and its intercept (c)
 $y = mx + c$

e.g. a line passing through (4,4) and (8,12) has passed through 8 units on the y axis and 4 on the x axis.

The slope is $8/4=2$, so $y = 2x + c$ and **$m=2$**

Since the line passes through (4,4), we have: $4 = 2 \times 4 + c$, so **$c = -4$**



Quadratics

Quadratics are a special case of function, they are a subset of polynomials

Polynomials are of the form: $y = ax^6 - bx^3 - cx + d$

They include multiple powers of x and possibly a constant too (think of this as dx^0)

Quadratics are one of the simplest polynomials, only going up to x^2

Solving quadratics

We try to 'solve' quadratics, by finding where they cross the x axis

The easiest way to do this is to factorise the expressions:

$$y = x^2 + 6x + 5 \quad \text{Expanded form}$$

$$(x + 5)(x + 1) \quad \text{Factorised form}$$

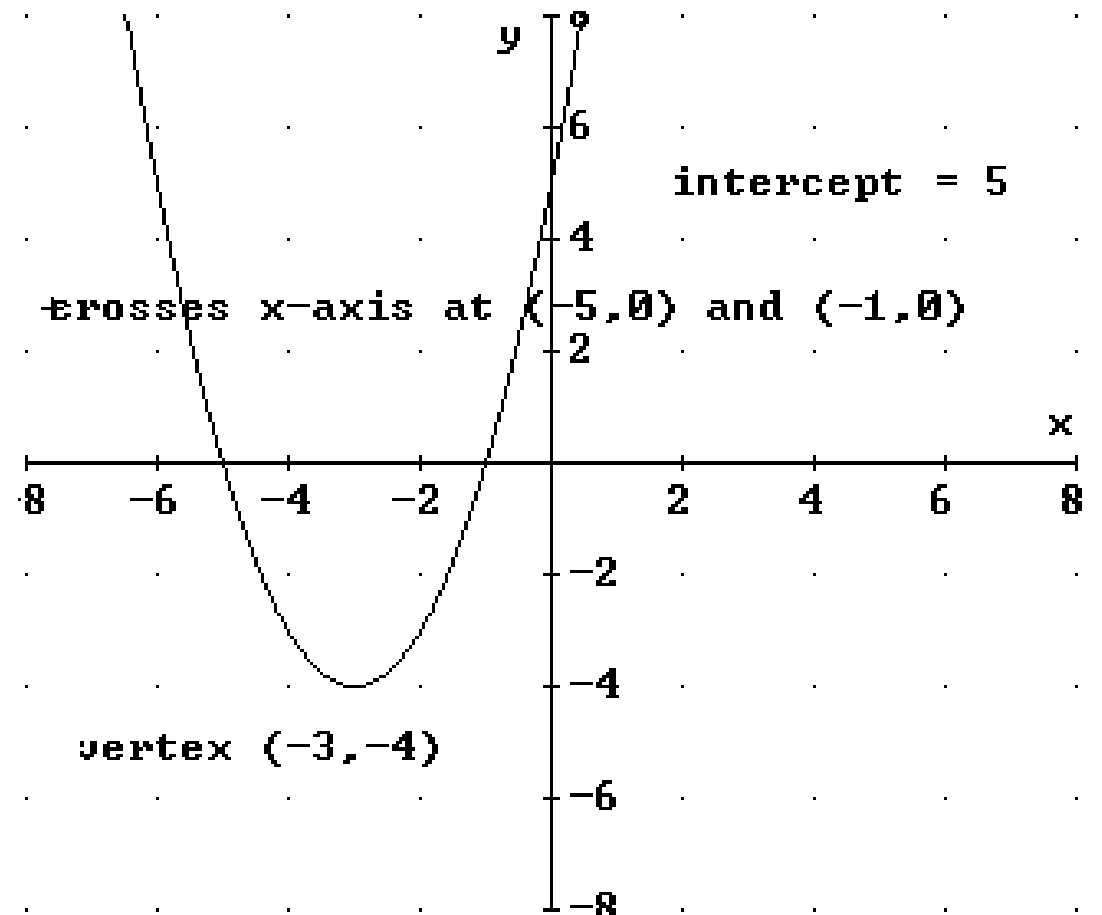
$$(x + 3)^2 - 4 \quad \text{Completed Square form}$$

Uses of forms

Completed square form: $(x + 3)^2 - 4$
gives us the position of the vertex

Factorised form: $(x + 5)(x + 1)$
gives us the points where the function
crosses the x axis

Expanded form: $y = x^2 + 6x + 5$
gives us the intercept



Completing the square

$$y = x^2 + 6x + 5$$

We ignore the constant for now and focus on the $x^2 + 6x$

We remove the square and halve the coefficient: $x + 3$

This is what we need to square to get our original expression: $(x + 3)^2$

Because we ignored the constant, we now need to fix that, $(x + 3)^2$ gives us $x^2 + 6x + 9$, so we need to remove 4 to get our original equation

$$(x + 3)^2 - 4 = x^2 + 6x + 5$$

Factorising

We looked at this before, and we tried to find numbers that added to give the coefficient and multiplied to give the constant:

$$y = x^2 + 6x + 5 \quad y = (x + 5)(x + 1)$$

It can be hard to do this, especially without simple integers, so we can use a formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Which works when:

$$y = ax^2 + bx + c$$

Factorising formula example

$$y = f(x) = x^2 + 3x + 2$$

So $a = 1, b = 3, c = 2$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

So

$$x = \frac{-3 \pm \sqrt{3^2 - 4 \times 1 \times 2}}{2 \times 1}$$

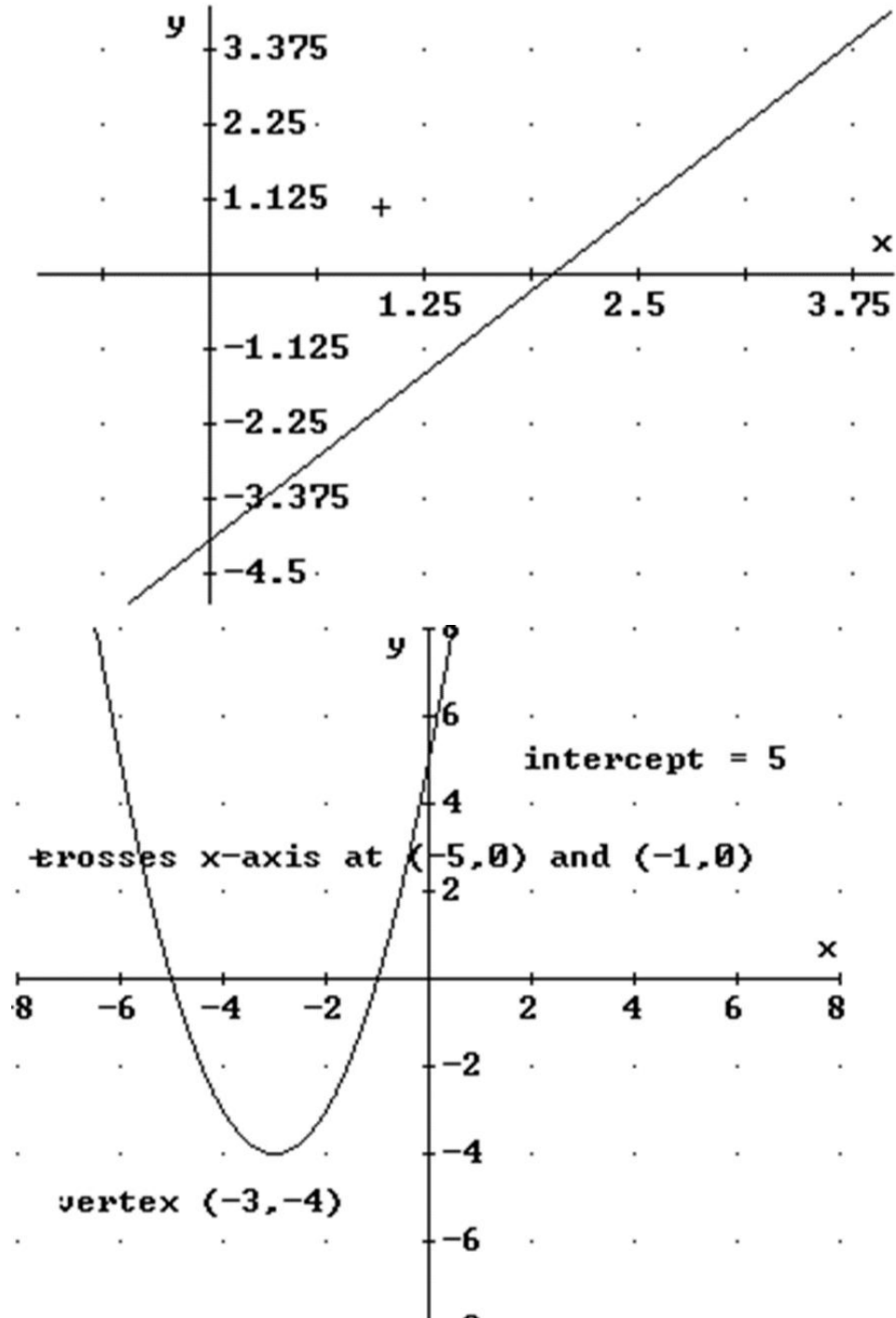
$$x = \frac{-3 \pm \sqrt{9 - 8}}{2} = \frac{-3 \pm 1}{2} = -1 \text{ or } -2$$

Key graph information

We have looked at finding the slope and intercept of straight lines

We have looked at finding the intercept, zeros and vertex of quadratics

I've been ignoring finding the slope of quadratics so far...

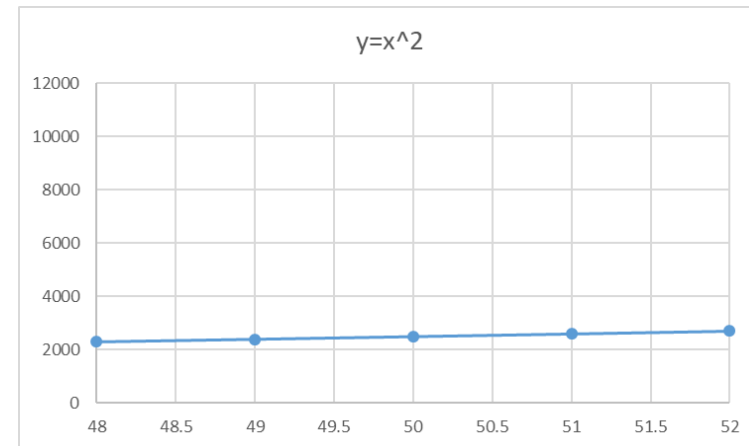
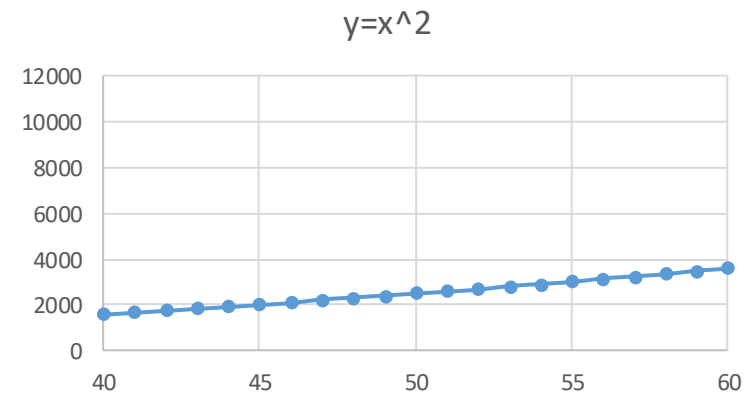
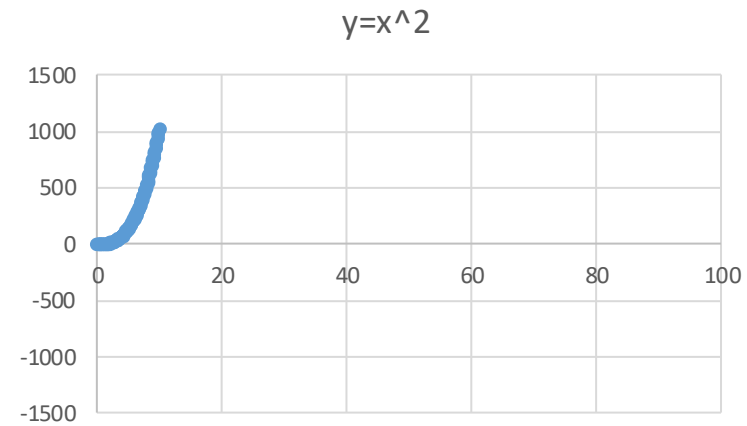


Tangent of curves

We can start to think of curves as being similar to straight lines, if we look at them on a small enough scale

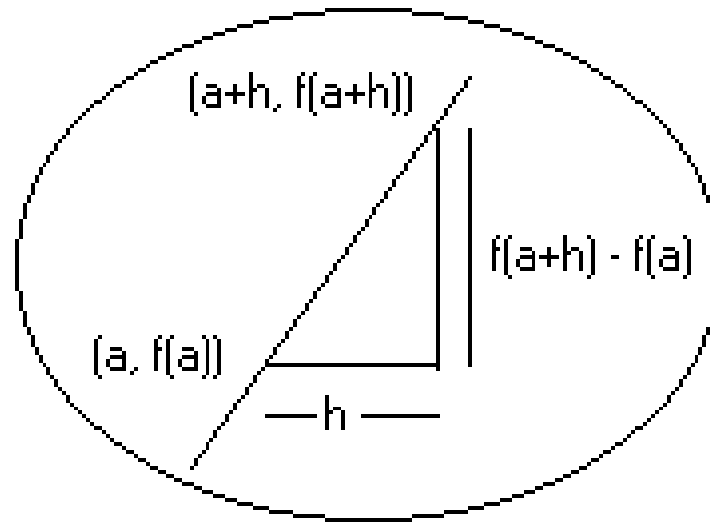
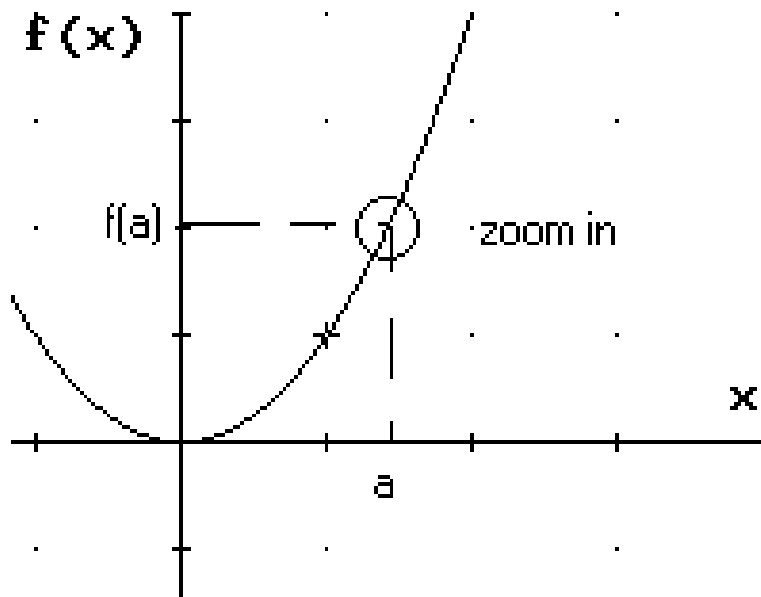
Here we zoom in on x^2

The slope that we see is 'roughly' correct, but **only** for that location!



Approximating gradient numerically

We can look at this gradient numerically – we'll have to approximate it, rather than getting a perfect answer

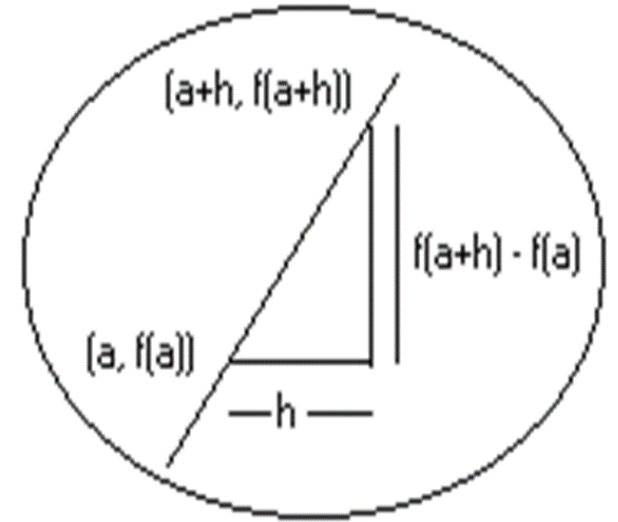


$$\hat{f}'(a) \approx \frac{f(a+h) - f(a)}{h}$$

Approximating gradient example

Find the slope of $f(x) = 2x^3 - 3x^2$ at the point (2,4)

We start by choosing a small step size: $h = 0.00001$



$$\hat{f}(a) \approx \frac{f(a+h) - f(a)}{h} = \frac{4.000120001 - 4}{0.00001} = 12.0001$$

As we use ever smaller values for h , we get closer to 12

Differentiation

Differentiation is the process of finding the slope of a function

It has two advantages to the numerical approximation we just looked at

- 1) It can tell us the slope at any point, not just at one location
- 2) It is an exact value, not an approximation (we make the step sizes infinitesimal)

notation

When you come across differentiation you may notice several notations used. They all have very different meanings.

$\frac{dy}{dx}$ or $\dot{f}(x)$ is used to show that something has been differentiated with respect to one variable, in this case x .

$\frac{\partial y}{\partial x}$ is used when something has been differentiated with respect to x , but the function is made up of more than one variable,

e.g. $y = 2x^2z + 3z$.

δy a small increment in the y -direction and δx a small increment in the x -direction.

differentiation

In general, if $y = ax^n$ then $\frac{dy}{dx} = anx^{n-1}$, where a and n are constants

The derivative of a constant is zero. e.g.

$$y = 2$$

$$y = 2x^0$$

$$\text{So } \frac{dy}{dx} = 0 \times 2x^{-1} = 0$$

Simple examples

$$y = x$$
$$\frac{dy}{dx} = 1x^{1-1} = 1$$

$$y = 2x + 3$$
$$\frac{dy}{dx} = 1 \times 2x^{1-1} + 0 \times 3x^{0-1} = 2$$

$$f(x) = x^3 + 3x$$
$$\dot{f}(x) = 3x^2 + 3$$

$$f(z) = 3z^3 - 2z^2 + z$$
$$\dot{f}(z) = 9z^2 - 4z + 1$$

medium examples

It is key, here, to remember the work we did on indices

$$y = \sqrt[4]{x} \text{ looks difficult, but: } \sqrt[4]{x} = x^{\frac{1}{4}} = x^{0.25}$$

$$\text{So: } \frac{dy}{dx} = 0.25x^{-0.75} = \frac{0.25}{x^{\frac{3}{4}}}$$

How about differentiating $x = (\frac{1}{3}) \times (y + 2)$?

$$\text{Rearranging gives } y = 3x - 2, \quad \text{so } \frac{dy}{dx} = 3$$

List of Standard derivatives

Function

y

Derivative

$\frac{dy}{dx}$

c (constant)

0

ax^n

anx^{n-1}

$\ln(x)$

$\frac{1}{x}$

ae^{kx}

ake^{kx}

$a \sin(bx)$

$ab \cos(bx)$

$a \cos(bx)$

$-ab \sin(bx)$

$\tan x$

$\sec^2 x$

Gaining answers for specific points

Velocity changes with time, $v = 3t^2 + 2$ we can differentiate this to get:

$$\frac{dv}{dt} = 6t$$

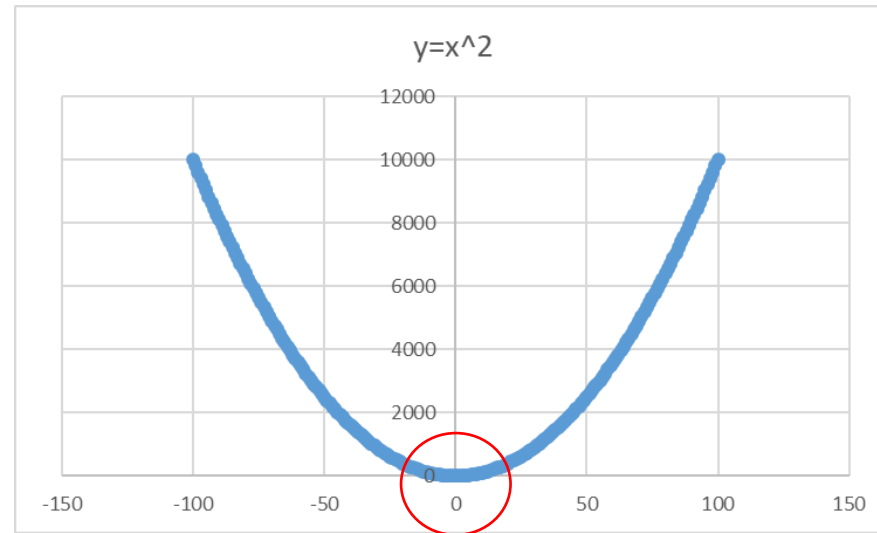
Now we can find the specific gradient at any point, e.g. $t=3$

$$\dot{f}(t) = 6 \times 3 = 18$$

In fact, differentiating velocity gives us acceleration, so we have found the acceleration at time $t=3$

Stationary points

There are places in some curves where the gradient is equal to zero



We looked at finding the vertex of a quadratic using the **completed square form** – we'd like to find a method that will also work for other, higher order polynomials, like:

$$y = 3x^4 - 2x^3 + 9x - 7$$

Try to put this into completed square from now

Cubic & Quartic

$$= \sqrt[3]{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)} + \sqrt{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3}$$

$$+ \sqrt[3]{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)} - \sqrt{\left(\frac{-b^3}{27a^3} + \frac{bc}{6a^2} - \frac{d}{2a}\right)^2 + \left(\frac{c}{3a} - \frac{b^2}{9a^2}\right)^3} - \frac{b}{3a}.$$

[illegible]

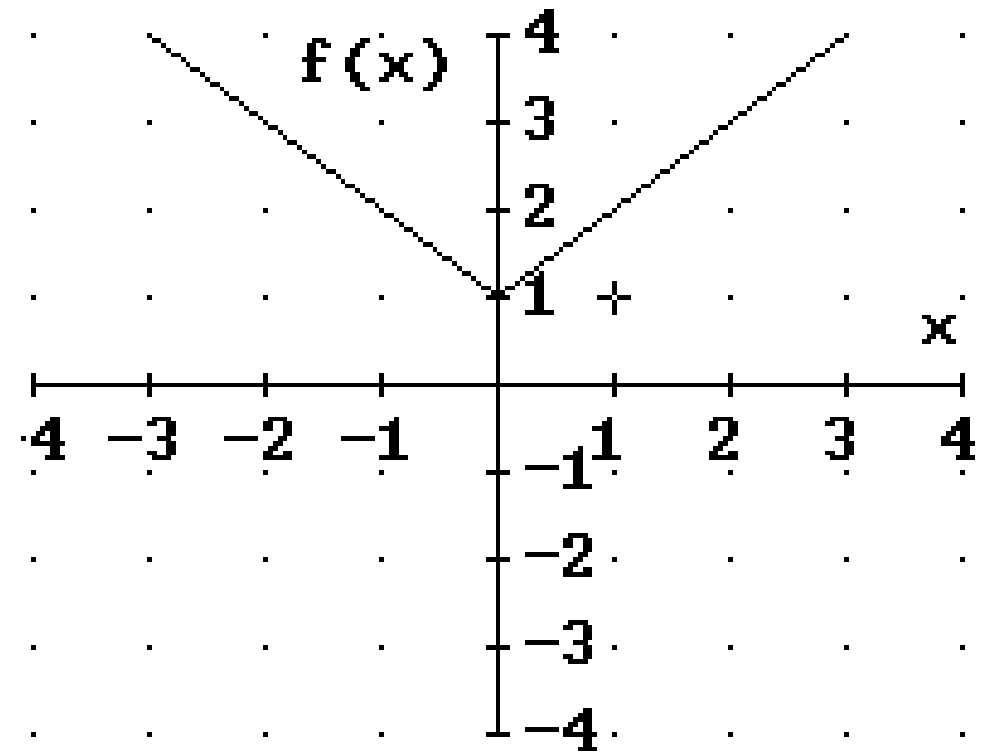
Stationary and turning points

Local maxima at $(0,3)$ it is both stationary and a turning point

Local minima at $(0,-3)$ it is both stationary and a turning point

Point of inflection at $(0,2)$ it is not a turning point but it is a stationary Point

Turning point at $(0,1)$ is not a stationary point



Finding stationary points

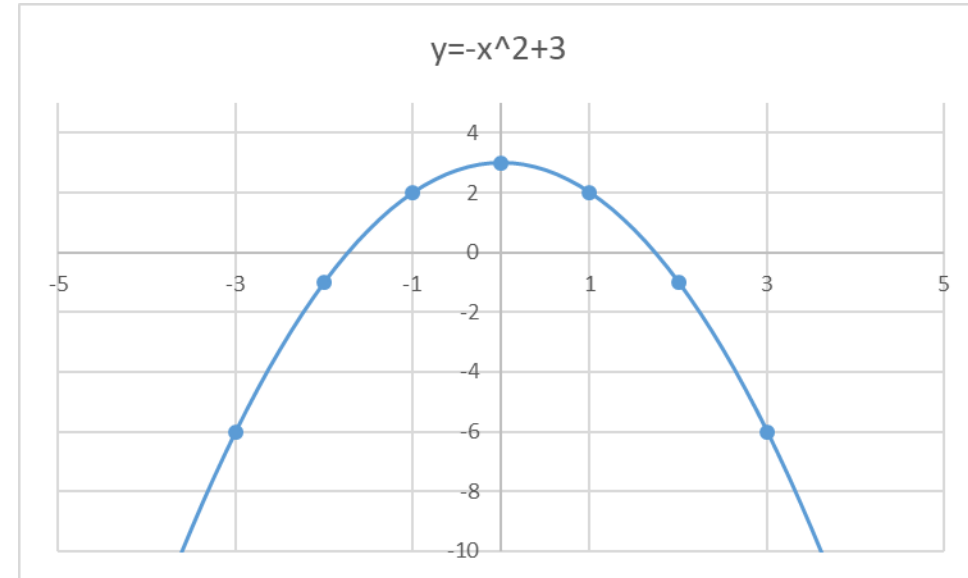
$$y = -x^2 + 3$$

We can find where this curve is zero by graphing it, but that's not always an option

We can also find the zeros by differentiating, since that tells us the slope of a curve, and setting the result equal to zero:

$$\frac{dy}{dx} = -2x$$

$$-2x = 0 \text{ only when } x = 0$$



Identifying stationary points

Consider: $y = x^3 + 2x^2 + x - 1$

The derivative is: $\frac{dy}{dx} = 3x^2 + 4x + 1$

Find this yourself using the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

We have: $x = -\frac{1}{3}$ or -1 , let's take $x = -1$ for this example

We know there is a stationary point here, but is it max, min or inflection?

Example continued

At $x = -1$ the gradient is zero

lets look to the left:

At $x = -1.1$ the gradient is given by $3(1.1)^2 + 4(1.1) + 1 = 0.23$

now look to the right:

At $x = -0.9$ the gradient is given by $3(0.9)^2 + 4(0.9) + 1 = -0.17$

The gradient is positive before the point and negative after, implying that this is a maxima

The tough stuff

So... the derivative of a function is the rate of change of the function

What about the derivative of the derivative? This is written: $\frac{d^2y}{dx^2}$

It tells us how the rate of change itself is changing!

This means that the first derivative tells what the gradient is and the second derivative can tell us if the gradient is increasing or decreasing

Final example

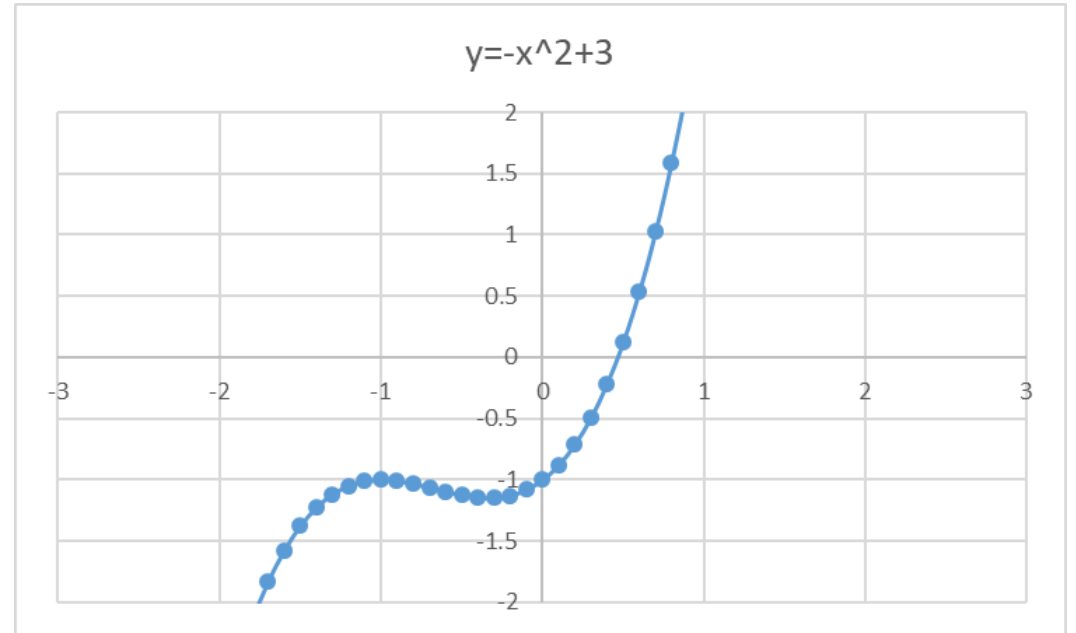
$$y = x^3 + 2x^2 + x - 1$$

The 1st derivative is: $\frac{dy}{dx} = 3x^2 + 4x + 1$

The 2nd derivative is: $\frac{d^2y}{dx^2} = 6x + 4$

If the second derivative is positive then we have a minima, if it is negative we have a maxima (if it is zero, we go back to basics!)

$$\frac{d^2y}{dx^2} = 6x + 4 = 6(-1) + 4 = -2$$



More help

Chapters 34 (gradients of curves) and 35 (techniques of differentiation, in 'Foundation Maths', by Tony Croft